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Homework #7

习题五 T19, 21, 22, 24, 25

T19 $S_X(\omega)=\frac{1}{(\omega^2+2)(\omega^2+3)}=\frac{1}{\omega^2+2}-\frac{1}{\omega^2+3}$

$=\frac{\sqrt{2}}{4}\frac{2\sqrt{2}}{\omega^2+2}-\frac{\sqrt{3}}{6}\frac{2\sqrt{3}}{\omega^2+3}$

$e^{-\alpha|\tau|}\xrightarrow{F}\frac{2\alpha}{\omega^2+\alpha^2}$

$S_X(\omega)\xrightarrow{F^{-1}}\frac{\sqrt{2}}{4}e^{-\sqrt{2}|\tau|}+\frac{\sqrt{3}}{6}e^{-\sqrt{3}|\tau|}$

即 $R_X(\tau)=\frac{\sqrt{2}}{4}e^{-\sqrt{2}|\tau|}-\frac{\sqrt{3}}{6}e^{-\sqrt{3}|\tau|}$

T21 (1) $\mu_X(t)=E[X(t)]=0$

$R_X(\tau)=E[X(t)X(t+\tau)]=E(A^2)\cos t\cos(t+\tau)+E(B^2)\sin t\sin(t+\tau)+E(C^2)$

$E(A^2)=E(B^2)=E(C^2)=\frac{1}{3}\Rightarrow R_X(\tau)=\frac{1}{3}\cos\tau+\frac{1}{3}$，与 τ 有关

故 $\{X(t),-\infty<t<\infty\}$ 为平稳过程

(2) $\langle X(t)\rangle=\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}X(t)\,dt=C$

$P\{\langle X(t)\rangle=\mu_X(t)\}\Rightarrow P\{C=0\}\neq 1$

故 $X(t)$ 的值不具有各态历经性

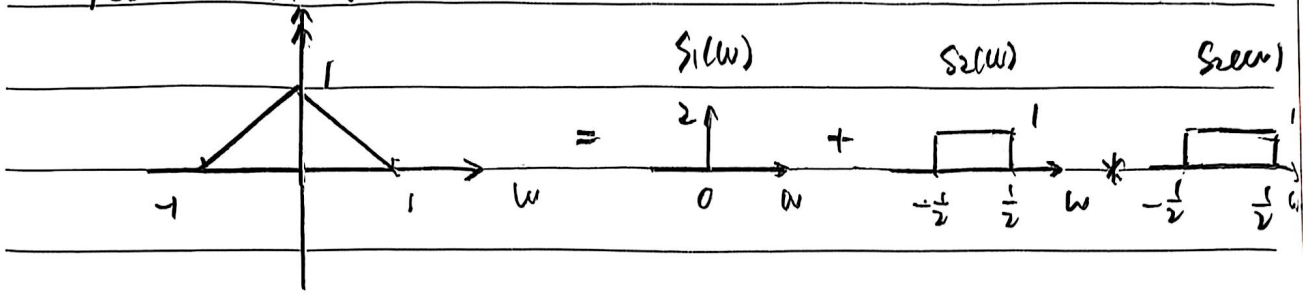
(3) $S_X(\omega)=\mathcal{F}(R_X(\tau))=\frac{\pi}{3}[\delta(\omega-1)+\delta(\omega+1)]+\frac{2\pi}{3}\delta(\omega)$

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T22 $S_X(\omega)$



② F ← $S_1(\omega)$ $\quad \frac{\sin\frac{1}{2}\tau}{\pi\tau}$ $\quad F$ → $S_2(\omega)$

故 $R_X(\tau) = \frac{2}{2\pi} + 2\pi\left(\frac{\sin\frac{1}{2}\tau}{2\pi}\right)^2 = \frac{1}{\pi} + \frac{2\sin^2\frac{1}{2}\tau}{\pi\tau^2}$

T24. 证明：(1) $Y(t) = X(t)\cos(t+\Theta)$

Θ	$\frac{\pi}{4}$	$-\frac{\pi}{4}$
P	0.5	0.5

$\mu_Y(t) = E[X(t)\cos(t+\Theta)]$

$= E[X(t)]E[\cos(t+\Theta)] = 0$

$R_Y(t) = E[Y(t)Y(t+\tau)]E[\cos(t+\Theta)\cos(t+\tau+\Theta)]$

$= \frac{1}{2}R_X(\tau)E[\cos\tau + \cos(2t+\tau+2\Theta)]$

$= \frac{1}{2}R_X(\tau)\cos\tau$

(2) $R_X(\tau)$ ← $S_X(\omega)$

$\cos\tau$ → $\pi[\delta(\omega-1)+\delta(\omega+1)]$

故 $S_Y(\omega) = \frac{1}{2\pi} \cdot \frac{1}{2} S_X(\omega) * \{\pi[\delta(\omega+1)+\delta(\omega+1)]\} = \frac{1}{4}[S_X(\omega-1) + S_X(\omega+1)]$

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15 取 $x(t)=e^{j\omega t}$，则 $y(t)=H(\omega)e^{j\omega t}$

$$H(\omega)e^{j\omega t}=e^{j\omega t}e^{j\omega L}-e^{j\omega t}$$

$$\Rightarrow H(\omega)=e^{j\omega L}-1$$

故 $S_Y(\omega)=|H(\omega)|^2 S_X(\omega)$

$$=(e^{j\omega L}-1)(e^{-j\omega L}-1)S_X(\omega)$$

$$=(2-2\cos\omega L)S_X(\omega)$$

$$=2S_X(\omega)(1-\cos\omega L)$$

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